Two functions on the natural numbers satisfying the same recurrences up to propositional equality are equal

The Univalent Foundations Program

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Editorial note. The counted statement is the author thm environment at source lines 103-124 and its proof environment opens on line 126, the line after the blank separator; the pairing is by adjacency and the retained proof range 126-132 is the whole environment.

Editorial note. The book label-number dump gives this label as 5.1.1 (env thm, chapter 5, section 1, first counter value), so the counters chapter 5, section 1 and thm 0 (zero-based) are set so that the item prints with the book number.

Editorial note. The counted statement and proof refer to no label outside the card: the retained ranges carry no cross reference command, and function extensionality and the induction principle are standard prerequisites rather than delegations of a step.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the uniqueness principle for functions on the natural numbers satisfying the same recurrences (author label thm:nat-uniq, book Theorem 5.1.1).

Allowed standard prerequisites (editorial).

- Induction on the natural numbers for a type family. A family over the natural numbers has a section as soon as it has an element over zero and a step over successors.. It is the induction that proves the pointwise equality of the two functions.
- Application preserves equality. Equal elements of the domain of a function have equal images, and a function applied to equal successors gives equal results.. It is what lets the inductive hypothesis be inserted inside the recurrence step.
- Function extensionality. Two functions are equal as soon as they agree on every argument.. It turns the pointwise equality obtained by induction into the equality of the two functions claimed by the statement.

Theorem 5.1.1. *Let $f, g : \prod_{(x:\mathbb{N})} E(x)$ be two functions which satisfy the recurrences*

$$e_z : E(0) \qquad \textit{and} \qquad e_s : \prod_{n:\mathbb{N}} E(n) \to E(\mathsf{succ}(n))$$

up to propositional equality, i.e., such that

$$f(0) = e_z \qquad \textit{and} \qquad g(0) = e_z$$

as well as

$$\prod_{n:\mathbb{N}} f(\mathsf{succ}(n)) = e_s(n, f(n)),$$
$$\prod_{n:\mathbb{N}} g(\mathsf{succ}(n)) = e_s(n, g(n)).$$

Then f and g are equal.

Proof. We use induction on the type family $D(x) :\equiv f(x) = g(x)$. For the base case, we have

$$f(0) = e_z = g(0).$$

For the inductive case, assume $n : \mathbb{N}$ such that $f(n) = g(n)$. Then

$$f(\mathsf{succ}(n)) = e_s(n, f(n)) = e_s(n, g(n)) = g(\mathsf{succ}(n)).$$

The first and last equality follow from the assumptions on f and g. The middle equality follows from the inductive hypothesis and the fact that application preserves equality. This gives us pointwise equality between f and g; invoking function extensionality finishes the proof. □